

Equivalent Expressions: Expanded & Factored Forms

Answer Key

Grade 6–7 Mathematics

Quick Answers

- | | |
|----------------|--|
| 1. $3x + 21$ | 7. $6x$ |
| 2. $10x + 15$ | 8. $6(3x + 4)$ |
| 3. $4(x + 3)$ | 9. (a) = 26, (b) = 26, — |
| 4. $5x + 7$ | 10. $2x + 5$ cm |
| 5. $7(2x + 3)$ | 11. $3x + 12$ |
| 6. $-4x + 24$ | 12. (a) = $9x + 18$, (b) = $9(x + 2)$ |
-

What is verified

11 of 12 problems fully machine-checked against SymPy.

— marks an answer only you can judge — problem 9. The worked solution states what a correct response must contain.

Curriculum

Level: Grade 6–7 Mathematics

7.EE.A – problems 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12

Difficulty 1–4 of 5 across 15 tagged checks.

- | | |
|---|---|
| 1. Expand: $3(x + 7)$
Solution: Distribute the 3 to both terms: $3 \cdot x + 3 \cdot 7$. | <div style="border: 1px solid black; padding: 2px; display: inline-block;">$3x + 21$</div> |
| 2. Expand: $5(2x + 3)$
Solution: $5 \cdot 2x = 10x$ and $5 \cdot 3 = 15$. | <div style="border: 1px solid black; padding: 2px; display: inline-block;">$10x + 15$</div> |
| 3. Factor out the greatest common factor: $4x + 12$
Solution: The GCF of 4 and 12 is 4: $4x = 4 \cdot x$ and $12 = 4 \cdot 3$. | <div style="border: 1px solid black; padding: 2px; display: inline-block;">$4(x + 3)$</div> |
| 4. Combine like terms: $8x + 2 - 3x + 5$
Solution: x -terms: $8x - 3x = 5x$. Constants: $2 + 5 = 7$. | <div style="border: 1px solid black; padding: 2px; display: inline-block;">$5x + 7$</div> |
| 5. Factor out the greatest common factor: $14x + 21$
Solution: The GCF of 14 and 21 is 7: $14x = 7 \cdot 2x$ and $21 = 7 \cdot 3$. | <div style="border: 1px solid black; padding: 2px; display: inline-block;">$7(2x + 3)$</div> |
| 6. Expand: $-4(x - 6)$ | |

Solution: Distribute -4 , watching signs: $-4 \cdot x = -4x$ and $-4 \cdot (-6) = +24$.

$$\boxed{-4x + 24}$$

7. Combine like terms: $2x + 9 + 4x - 9$

Solution: x -terms: $2x + 4x = 6x$. Constants: $9 - 9 = 0$, so no constant remains.

$$\boxed{6x}$$

8. Factor out the greatest common factor: $18x + 24$

Solution: The GCF of 18 and 24 is 6: $18x = 6 \cdot 3x$ and $24 = 6 \cdot 4$.

$$\boxed{6(3x + 4)}$$

9. Consider $2(3x + 1)$ and $6x + 2$. (a) evaluate the first at $x = 4$; (b) evaluate the second at $x = 4$; (c) equivalent for every x ?

Solution: (a) $2(3 \cdot 4 + 1) = 2 \cdot 13 = 26$.

$$\boxed{26}$$

(b) $6 \cdot 4 + 2 = 24 + 2 = 26$.

$$\boxed{26}$$

(c) Model answer: yes — expanding gives $2(3x+1) = 6x+2$ exactly, so the two expressions agree for *every* x , not just at $x = 4$. (Instructor judges the sentence; a matching test value alone is evidence, and the expansion is the proof.)

10. A banner has area $(12x + 30)$ square centimeters and height 6 cm. Write and simplify an expression for its width.

Solution: Width is area divided by height: $\frac{12x + 30}{6}$. Split the division over both terms:

$$12x \div 6 = 2x \text{ and } 30 \div 6 = 5.$$

$$\boxed{2x + 5 \text{ cm}}$$

11. Simplify completely: $4(2x + 3) - 5x$

Solution: Expand first: $4(2x + 3) = 8x + 12$. Then combine like terms: $8x + 12 - 5x = (8x - 5x) + 12$.

$$\boxed{3x + 12}$$

12. Challenge. Consider $6(x + 2) + 3x + 6$. (a) simplest expanded form; (b) factor the result.

Solution: (a) Expand: $6(x+2) = 6x+12$. Combine like terms: $6x+12+3x+6 = 9x+18$.

$$\boxed{9x + 18}$$

(b) The GCF of 9 and 18 is 9: $9x = 9 \cdot x$ and $18 = 9 \cdot 2$.

$$\boxed{9(x + 2)}$$